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Probabilistic stability analysis: the way forward for stability analysis of sustainable power systems

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Future power systems will be significantly different compared with their present states. They will be characterized by an unprecedented mix of a wide range of electricity generation and transmission technologies, as well as responsive and highly flexible demand and storage devices with significant temporal and spatial uncertainty. The importance of probabilistic approaches towards power system stability analysis, as a subsection of power system studies routinely carried out by power system operators, has been highlighted in previous research. However, it may not be feasible (or even possible) to accurately model all of the uncertainties that exist within a power system. This paper describes for the first time an integral approach to probabilistic stability analysis of power systems, including small and large angular stability and frequency stability. It provides guidance for handling uncertainties in power system stability studies and some illustrative examples of the most recent results of probabilistic stability analysis of uncertain power systems.

This article is part of the themed issue 'Energy management: flexibility, risk and optimization'.

1. Introduction

The function of a power system is to generate electric energy economically, with the minimum ecological disturbance, and to transfer this energy over transmission lines and distribution networks with the maximum efficiency and reliability for delivery to customers at virtually fixed voltage and frequency. Electric power systems, in comparison with other man-made systems

(for example, communications, gas or water networks), are probably the most expensive in terms of capital invested and the most influential in terms of the disruption to our mode of life in case of breakdown. Furthermore, they are probably the most visually intrusive in terms of the impact on the landscape (such as power stations, cooling towers and transmission lines) and the most ecologically intrusive in terms of the thermal, chemical and potential radiological pollution. For the largest part of their existence, since being conceived close to 150 years ago, they have been designed and operated as large (typically state-owned) monopolies. The power utilities were *vertically integrated*, meaning they owned and operated the entire value chain, from the power plant to the electricity meter at the end-user facility. The start of the deregulation of electric power systems began in the late 1980s and early 1990s and is combined with the accelerated proliferation of new types of generation (different from synchronous generators (SGs)) in the early 2000s as well as the constant growth in power electronics (PE) interface-connected loads since the 1960s. As a consequence, the nature and operational principles of power systems have changed, and are still changing, substantially.

(a) Transformative changes of power systems

The electrical power systems of the future will be substantially different compared with their present states in terms of design, constituent components and operation. They are already, and will be even more so in the future, characterized by an unprecedented mix of a wide range of electricity-generating technologies (e.g. fossil fuel, gas, nuclear, hydro, wind, solar and tidal), efficient electrical power transmission-enabling technologies (e.g. high-voltage direct current (HVDC) and flexible AC transmission system (FACTS) devices), as well as responsive and highly flexible demand (e.g. PE-interfaced ‘intelligent’ appliances, electric vehicles (EVs)) and storage technologies (e.g. batteries, supercapacitors, flywheels, hydrogen and superconducting magnetic energy storage) with significant temporal and spatial uncertainty. These evolved power systems will be principally characterized by the following:

- (i) Evolving and new market structures and operation with a liberalized energy market and increased cross-border (between countries and transmission system operators) bulk power transfers to maximize the effectiveness of market mechanisms.
- (ii) New generation and storage technologies, mostly PE-interfaced and often not visible to the system operator (less than 100 MW typically invisible to the system operator), including mainly large onshore and offshore wind farms and grid-connected solar technologies, thermal and photovoltaics (PV), as well as small-scale widely dispersed technologies in distribution networks (mostly PV). These renewable energy sources (RESs) have a low energy density and are therefore spatially distributed. They are connected to the network through PE interfaces. Some of them are intermittent and/or stochastic (dependent on weather and meteorological conditions). Intermittent and stochastic RESs thus provide little control and cause much uncertainty in operation. Because of the intermittency and stochasticity and the consequent sub-maximal utilization due to varying loading factor (e.g. wind farm loading factor broadly varies between 25% and 35%), the renewable generation capacity will have to represent a significantly larger fraction of the total installed capacity and be supported by energy storage (to compensate the lack of generation from RESs at certain times) to ensure continuity and security of supply as well as provision of voltage and frequency support to the network if required.
- (iii) Proliferation of PE-based ‘efficient transmission facilitating’ technologies, including increased use of HVDC lines of both line commutated converter (LCC) and predominantly voltage source converter (VSC) technology with increasing use of multi-modal converters (MMC) in both meshed AC networks and as a DC supergrid. Furthermore, there will be an increased presence of static and active (PE-based) shunt and series compensation of long and short AC transmission lines, as well as increased

deployment of FACTS devices to improve the controllability and flexibility of the existing transmission network.

- (iv) New types and different operational patterns of load with typically greater flexibility connected at customer premises, including heat pumps, PE-interfaced loads, efficient lighting as well as growing use of EVs characterized by spatial and temporal uncertainty.
- (v) Increased monitoring at all voltage levels and acquisition of a large amount of data. The data are typically multidimensional, multiscale, spatially distributed in the form of either time series (with sampling rates ranging from milliseconds to hours) or event-triggered, often incomplete or noisy due to sensing and/or communication problems.
- (vi) Tendency towards, and requirement for, increased efficient energy supply resulting in increased consideration of different energy carriers (multi-energy networks).
- (vii) Requirement for increased information security (cyber security) due to a wide range of integrated diverse 'intelligent' PE devices and information and communication technologies (ICTs).

(b) Increased uncertainties in power systems

One of the main attributes of the above systems is the increase in uncertainties associated with system operation and modelling. There are generally two forms of uncertainty associated with any system modelling and analysis: (i) aleatory uncertainty (irreducible uncertainty and variability), which represents the inherent random behaviour of a system commonly modelled by probabilistic distribution functions and propagated by probability-based approaches (sampling, analytical methods, probabilistic chaos expansion); (ii) epistemic uncertainty (reducible uncertainty and state of knowledge uncertainty), which models the uncertainty in parameter estimation due to data shortages or model simplification.

The major sources of uncertainties in power systems grouped based on the system components from which they originate are identified below:

(i) Network-based uncertainties

- Network topology.
- Network parameters and settings (e.g. settings of tap-changing transformers, temperature-dependent line ratings, line and cable parameters).
- Network observability and controllability.

(ii) Generation-based uncertainties

- Generation pattern and mix (size, output of generators, types and location of generators, i.e. conventional, renewable, storage, PV generation at the distribution level).
- Output uncertainty of renewable generation due to forecasting errors.
- Models and parameters (conventional and renewable generation and storage technologies, individual and aggregate models of wind farms and distributed PV generation).

(iii) Load-based uncertainties

- Time and spatial variation of load (e.g. location of EVs in particular).
- Load composition (mix).
- Load models and parameters, both frequency and voltage dependence of load, including conventional types of loads and rising number of new PE-interfaced loads and efficient lighting.
- Load forecasting uncertainty.

(iv) System controls-based uncertainties

- Parameters of generator controllers (excitation and voltage controllers, governors, damping controllers, PE interface controllers).

- Parameters of network controllers (secondary voltage controllers, controllers of FACTS devices and HVDC controllers).

(v) Operating condition-related uncertainties

- Contractual power flows as a consequence of different market mechanisms and price.
- System faults (type, location, duration, frequency, distribution, impedance of system faults and disturbances).

(vi) ICT-related uncertainties

- Noise, measurement errors, time delays, loss of signals, bandwidth, missing data.

(vii) Weather/climate-related uncertainties

- Wind speed, wind direction, temperature, humidity, solar irradiation, tidal/wave conditions.

Each form of the uncertainty listed above has a specific representation and quantification method. Therefore, the selection of the appropriate method to model relevant uncertainty comes after the identification of the type of the uncertainty. For some of these uncertainties, it may be possible to produce a sufficiently accurate model based on historical values or data tolerance values from manufacturers. For some parameters, however, the level of uncertainty may be unquantifiable without additional monitoring, measurement or analytical effort. More importantly, though, it may not even be necessary to accurately model all uncertainties in the system, as many may have little impact on the system phenomena of interest, despite adding considerable computational burden when modelled more accurately. It is important to mention that, in spite of the trend of increasing monitoring in power systems in general, the transmission companies still have little experience in collecting standardized network performance data such as outage rates and repair times.

(c) Key challenges in stability studies of future power systems

Considering the above structure of future power systems and increased levels of uncertainties, the key research challenges for stability analysis of future power systems can be grouped into three broad areas:

- (i) Efficient use of and reliance on existing and newly acquired data through deployed local measurement devices and two-way communications-enabled meters and global wide-area monitoring systems (WAMS) for state estimation, static and dynamic equivalents and control (including real-time control). This implies efficient data management (signal processing, aggregation and transmission), reliability of, and reliance on, ICT networks, as they are essential for both static and dynamic observability of the system.
- (ii) Modelling for steady-state and dynamic studies of
 - Large interconnected networks with generation mix, FACTS devices and short- and long-distance bulk power transfers using HVDC lines.
 - Clusters of RESs (generation and storage) of the same or different type.
 - Parts of, or the whole, LV and MV distribution network with thousands of integrated RESs.
 - Demand, including new types of energy-efficient and PE-controlled loads, heat pumps, customer participation and behavioural patterns, EV, etc.
 - Centralized and distributed storage technologies (i.e. virtual storage plant) for provision of ancillary services.
 - Interconnected critical infrastructure systems, 'system of systems' and, as a subset of these, networks with different energy carriers and self-sufficient energy modules/cells.

(iii) Design of advanced controllers and control structures including

- Design of supplementary controllers based on WAMS to control, including real-time control, and stabilize a large system or parts of the system (the structure and extent of which may vary depending on connected technologies and the aims of control) with uncertain power transfers and load models and stochastically varying and intermittent PE-connected generation, demand and storage, i.e. reduced inertia systems.
- Design of new risk-limiting control systems/structure, including consensus, distributed or hierarchical control, adaptive and close to real-time control for power networks with fully integrated sensing, ICT technologies and protection systems.

Considering the evolving power system with increased uncertainties and abundance, and growing measurement data and challenges identified above, the key question that needs answering is: *Are the tools currently in use for system modelling, analysis and control adequate, and if not, how should we modify them, or what other tools should we be using?*

This paper addresses the above question by proposing a probabilistic approach to power system stability studies. It presents an integral approach to probabilistic stability analysis of power systems, including small and large angular stability and frequency stability, and provides guidance for handling uncertainties in power system stability studies and some illustrative examples of the most recent results of probabilistic stability analysis of uncertain power systems.

2. Probabilistic approach to power system stability studies

In general, a deterministic approach to power system analysis evaluates the system performance based on a specific scenario and ignores the uncertainties in the system states and parameters. It assumes that all the states are known and constant. To ensure the robustness of the analysis, a range of characteristic scenarios with fixed parameters is considered. Owing to the potentially very large number of scenarios, in particular in large power system studies, in many cases only the worst-case scenario (which cannot be objectively defined) is analysed and corresponding operational decisions made to ensure system integrity and stability. The system stability in deterministic studies is treated as binary (a certain condition is either stable or unstable), which does not take into account various uncertainties associated with controllers' gains, production of unobserved distributed generation, etc. When performing deterministic analysis, the selection of the parameters and operating conditions is very important, as it can lead to the problem considered being either underestimated or overestimated; hence the decision made might be suboptimal.

A probabilistic approach, on the other hand, considers the probability distribution for one, some or all of the uncertain parameters, and can therefore better reflect the actual system behaviour. It also provides answers to questions like 'What could the output be?' and 'How likely is that to happen?', i.e. it can appropriately determine the risks to which a given operating condition is exposed. Furthermore, it can determine how sensitive the output is to variation in the input parameters. Even though it has been long recognized that deterministic studies may not adequately represent the full extent of system dynamic behaviour, the probabilistic approach has not been widely used in the past in power system studies except in some specific studies mainly related to system reliability. Probabilistic methods are particularly suited for the analysis of a system with randomness and uncertainty, which are clearly key characteristics of future power systems. Appropriate accounting for uncertainties in system parameters and stochastic variability in parameters, operating conditions, disturbances and other variables affecting system performance, e.g. weather conditions, can provide a much more realistic picture of expected system performance. This can facilitate a much better understanding and more accurate prediction of the system static and dynamic behaviour and hence deployment of appropriate mitigation actions.

Some of the advantages of a probabilistic approach include the following:

- The use of deterministic studies based on *worst-case* scenarios restricts the efficient utilization of the system capacity, while a probabilistic analysis allows increased utilization of existing assets.
- Deterministic planning typically uses a limited number of user-specified contingencies to highlight the extreme operating conditions which may not adequately reflect system operation and associated risks. The accuracy of this assessment is reliant on the correct selection of representative conditions (which may not be trivial or guaranteed for systems with a large number of parameters and high degrees of variability of these parameters).
- In deterministic studies, the variation of generation and/or load profile is usually presented considering discrete values (e.g. high, medium and low levels), while in probabilistic planning these parameters can present a wide variety of scenarios considering hourly, daily, seasonal, and annual patterns and as such providing a much more accurate estimate of true system conditions.

Despite these clear advantages, the single largest being the increased data and greater understanding of true system performance that probabilistic analysis allows, there are a number of challenges which have slowed the adoption of probabilistic approaches and can affect the accuracy of the analysis. These include the following:

- (i) The need for detailed understanding of how different uncertainties are distributed. The results of probabilistic analysis are dependent on the validity of the input data. If the uncertainties are not modelled correctly, then the results should not be expected to be accurate or meaningful. With no history of extensive monitoring or probabilistic modelling in power systems, there is a shortfall in the knowledge about power system uncertainties and their distributions.
- (ii) The need for appropriate representation of the correlations between different uncertainties and the different interactions between system parameters. As in the point above, these interdependences can have a significant impact on the results.
- (iii) The need to identify the most influential uncertain parameters that will have the largest impact on the accuracy of the results. The computational burden associated with modelling all power system uncertainties (particularly in large interconnected systems) is extremely large. Reducing this burden is critical to ensuring the uptake of probabilistic methods. This can be achieved by identifying the most critical parameters and therefore neglecting those with limited impact.
- (iv) Finally, the areas of probabilistic analysis and modelling of uncertainties have not typically been part of the curriculum in power system training and education, and therefore are not something that power system engineers are generally familiar with or accustomed to use.

(a) Modelling of system uncertainties

The uncertainties in any system parameters or operating conditions can be generally categorized as random and non-random. Random uncertainties are repeatable and have known probability distributions, e.g. load, generation costs, availability of generators. Non-random uncertainties are not repeatable and, hence, their statistics cannot be derived from past observations [1,2]. They can be considered in system studies either using scenario generation techniques, where multiple alternative scenarios are considered, or using a stochastic formulation approach, where a mathematical model is used to represent the uncertainty. In selecting the appropriate modelling approach and probability distribution to model relevant parameter uncertainty in general, the

Table 1. Probability distributions of system input parameters.

variables	probability distributions
<i>parameter</i>	
power generation	normal [3,4], discrete [5]
wind speed/power	Weibull [3,6–8], normal [9], discrete normal [10], joint Gaussian [11], lognormal [8], gamma [8]
solar power	Weibull [6], PDF of historical data [12], beta [13]
power system load	normal [3,4,6], Gumbel [7], discrete normal [10], PDF of data [12], joint normal [5]
<i>disturbances</i>	
fault occurrence	Poisson [14–16], binomial [16]
fault type	discrete [4,7,17], PDF of historical data [17]
fault location	discrete [7], uniform [4,14], PDF of historical data [16]
fault impedance	normal [4,7]
fault-clearing time	normal [4,7,15], Poisson [14]
fault duration	Rayleigh [4,7]
contingency/failure	Markov model [10]
time to failure	exponential [18]

following practice should be followed:

- (i) Parameters that grow linearly (e.g. errors, offsets) can be modelled as following a normal (Gaussian) probability distribution.
- (ii) Parameters that grow exponentially (e.g. prices, income and population) can be modelled as following a lognormal probability distribution.
- (iii) Uniformly distributed parameters over a region can be modelled using either discrete uniform or continuous uniform distributions.
- (iv) Discrete binary (yes/no) events with a given probability can be modelled using a Bernoulli or binomial distribution.
- (v) Events with k outcomes, each with a given probability, can be modelled using a multinomial distribution (an extension of the binomial distribution).
- (vi) Events occurring independently at a given rate can be modelled using Poisson, exponential or gamma probability distributions, depending on the output being modelled (for example, the number of events in a given time period or the time until the next event).

Table 1 summarizes different power system parameters and disturbances, with corresponding probability distributions used in past power system stability studies. Illustrative references are also given for further reading.

An important aspect of modelling uncertainty in system parameters is to consider possible correlations between different parameters. Dependence of uncertain parameters on each other can lead to a major impact on the aggregated uncertainty, and these features have been discussed and modelled extensively in past studies, e.g. [19–21]. The dependence of multiple variables is typically modelled using copula theory. A *copula* (Latin for ‘a link, tie, bond’) is a multivariate probability distribution for which the marginal probability distribution of each random variable is uniform. Copulas are used to describe the dependence between random variables [22]. In the consideration of correlated inputs into a probabilistic power system model, three types of uncertainty correlation have been discussed in the past: (i) correlation among power sources,

e.g. wind power plants and PV arrays [19]; (ii) correlation among loads [20]; and (iii) correlation between power sources and load [21].

(b) Probabilistic stability analysis approaches applied in power system studies

Though power system studies, in general, and stability studies, in particular, traditionally have not been performed using probabilistic approaches, there has been some work in this area reported in the past. Some of the most widely used probabilistic methods in the past in power system studies in general (including power system stability studies) are briefly discussed below. The techniques most widely used include Monte Carlo (MC) simulation-based methods (including sequential MC and quasi-MC (QMC)), point estimate (PE), cumulant-based, probabilistic collocation, probabilistic game theory and unscented transformation. For some of these main advantages, limitations and areas of applications are given below.

(i) Monte Carlo simulation-based methods

MC simulation, e.g. [23], involves repeated random sampling of system uncertainties in order to obtain a large dataset (i.e. a numerical solution) from which the distribution of an unknown probabilistic entity, i.e. output probability density function (PDF) can be determined. The MC method is very flexible and virtually limitless for analysis and the algorithms can be easily extended and developed. Repeated numerical simulation, however, can take a very long time depending on the complexity of the system, the number of parameters modelled as uncertain, the type of study performed and (of course) the computational power exploited. As MC simulation is based on computationally pseudo-random selection of parameters from the search space, a very large number of simulations (samples) is required for adequate coverage of the search space and reasonably accurate estimation of the possible outcomes. Even for a large number of simulations, the uniformity of sampling in different search directions in the search space is not guaranteed. For a large complex system with numerous parameters, the simulations can be prohibitively computationally expensive and not practicable.

A traditional MC simulation assumes that each sample is a unique point in time that is not correlated to all other points. Each sample is independent. Sequential MC simulation models, on the other hand, consider time-based dependence, i.e. that an action or decision now may impact the following time period being studied. They are typically used to analyse outages, system restoration processes and the resulting impact on the system that these events have [24]. The sequential MC method can perform accurate frequency and duration assessments of different events, which is a clear advantage compared with the conventional MC method, which assumes time independence of events and parameters. The sampling is still random and the uniformity of sampling in different directions is not guaranteed. It also requires very high computational effort, which limits its application for a large complex system [25].

QMC methods [3,26,27] are in a way similar to the conventional MC method. While the conventional MC method generates a pseudo-random sequence when sampling parameters, QMC generates quasi-random, low-discrepancy sequences such as Halton or Sobol sequences. Quasi-random sequences have a more uniform behaviour within the search space and are based on equally distributed sequences in different search directions. This means that the sample points selected by a QMC approach will fill the search space more uniformly and that acceptable accuracy can be reached with significantly fewer samples.

(ii) Point estimate methods

PE methods [23,28] use a small number of specified point values to represent the distributions of any system uncertainties. The point values for different uncertainties are combined in different permutations to form *concentrations*. The system model is then evaluated for these different concentrations and the output values are combined with concentration weightings to estimate the moments of the output distribution. Different PE methods use different numbers of points. They

are generally much more computationally efficient than MC-based methods; e.g. compared with conventional MC methods, a two-PE method requires only $2n$ samples for n uncertain parameters, which results in significant reduction in time and computational effort. The PE approaches can be used with symmetric or asymmetric variables but not with correlated system uncertainties. Furthermore, the type of PE method that can be used is dependent on the distribution of input uncertainties, and certain conditions must be met to ensure that real (non-complex) sample points are generated. Further details on this can be found in [29].

(iii) Cumulant-based methods

A cumulant-based method [23,30] uses the cumulants of probability distributions to derive an analytical, or semi-analytical, determination of the system output distribution. A cumulant is a statistical measure of a distribution, i.e. an alternative to the moment of the distribution. The input cumulants are mapped to the output cumulants through sensitivity functions describing the input–output behaviour. These methods are extremely dependent on the accuracy of this relationship and are often only valid when the uncertainty is small and the input–output relationship is close to linear. They are typically characterized by reduced computational burden and complexity [31] and they can avoid convolution calculations in probabilistic power flow studies [30]. They are, however, extremely dependent on the correct derivation of the input–output sensitivity.

(iv) Probabilistic collocation method

The probabilistic collocation method (PCM) [23,32–34] expresses the system model output as a polynomial function of the uncertain parameter set. The basic idea is to use a small number of sample points to create a computationally cheap function that can be used to replace the high burden of the full power system study during further repeated sampling. Polynomial functions of increasing order and complexity can be used to capture high-order interactions, albeit at greater computational cost. Uncertainties typically must be pre-ranked to identify critical parameters, as the number of samples required to produce the function grows exponentially with the number of considered uncertain variables. This method is very computationally efficient provided the number of considered uncertainties is small, or reduced to a small number [33,35], and requires significantly reduced computational time compared with the conventional MC method [36].

3. Examples of probabilistic power system stability studies

(a) Test system

The test system used to illustrate probabilistic stability analysis is a modified 68-bus IEEE NETS–NYPs test system (New England Test System–New York Power System). The system, shown in figure 1, consists of five interconnected areas, 16 conventional SGs, seven RESs and 34 loads.

The parameters for 16 SGs in the network are selected to represent different types of electric power plants found in typical power systems. Out of the total of 16 SGs in the network, three are modelled using parameters for typical SGs found in nuclear power plants, two using parameters for typical hydro generators and 11 using parameters for generators found in steam or gas-fired power plants. All round rotor SGs (e.g. those found in nuclear, steam and gas-fired power plants) are represented by a full sixth-order dynamic model, while the fifth-order dynamic model was used to represent hydro generators. All generators, except G9, are equipped with slow IEEE DC1A-type exciters. G9 is equipped with a fast-acting static exciter of the IEEE STI type and with a power system stabilizer (PSS). All generators are also equipped with generic governors typically found in corresponding power plants: generator G1 with a GAST speed governor, G3 and G9 with an IEEE G3 governor (hydro turbine) and G2–G8 and G10–G16 with an IEEE G1 governor (steam turbine).

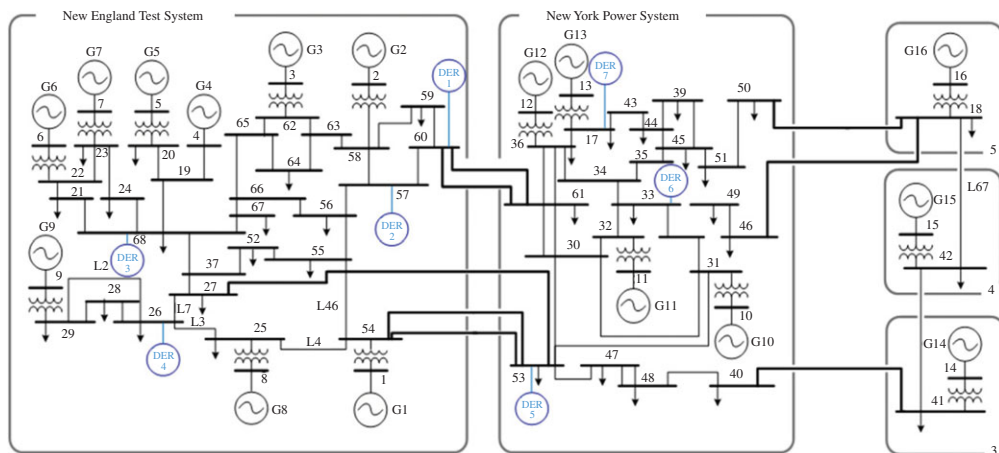


Figure 1. Modified IEEE NETS–NYPS test system. (Online version in colour.)

Out of the seven renewable generators modelled in the network, four are modelled as wind plants and three as a combination of wind and PV plants. The WTG Type 3 generator model, doubly fed induction generator (DFIG), is used to represent wind generators, and the WTG Type 4 generator model, full converter connected (FCC), to represent both wind generators and PV units. The modelling approach is similar to the one recommended by two recent working groups: WECC [37] and IEC [38]. The DFIG model takes into consideration the aerodynamic part and the drive train, the mechanical side of the converter. The model also includes the pitch control of the turbine blades. The electrical controls that define the control of the active and reactive power of the unit are modelled appropriately. The induction generator model incorporates the rotor flux transients but neglects the stator flux transients, which is common practice in power stability analysis [38]. The rotor side converter protection (over and under turbine speed, over and under terminal voltage limits) is also modelled by representing the crowbar system. The WTG Type 4 model represents FCC wind generators, and any generator connected to the grid through a full converter interface, e.g. PVs, can be represented as the Type 4 WTG model for system stability studies [37] because the converter decouples the mechanical dynamics of the unit from the electric grid [39,40]. The model employed within these studies is the WTG Type 4 generic model available in DiGSILENT PowerFactory developed using a similar approach as discussed in [36,37]. A current controller, real (active)–reactive power PQ controller and over frequency power reduction control of the converter are included in the model. Further details about the system and references to the sources of data used in simulations can be found in [41–46] and appropriate system settings and computational tools downloaded from <https://data.mendeley.com/>.

In deterministic simulation, the real power output and the nominal apparent power of each synchronous generator are fixed (but different for different case studies). In probabilistic simulation, the nominal apparent power of each synchronous generator is fixed; however, the real power output is dependent on the optimal power flow (OPF) calculation following allocation of available renewable generation and depending on system loading.

Transmission lines are modelled using the conventional π equivalent model with lumped parameters and the transformers are modelled by short-circuit reactance. All system loads are modelled using the conventional constant-power static load model.

In the probabilistic simulations presented in this paper the following process has been followed. The probability distributions of uncertain system parameters are generated in Matlab. Matpower 5.1 is then used for solving OPF for each set of different system parameters and operating conditions. The output results (probability distributions of parameters and corresponding OPF-generated generator scheduling) are stored in appropriate files and the DiGSILENT PowerFactory software is run (automatically, using Matlab ‘master’ programme) the

required number of times (1000 times) to produce relevant results for different types of system stability analysis. The output results from each run of PowerFactory are stored and then analysed and illustrated off-line using Matlab.

(b) Case studies

The six case studies, listed below, are used to illustrate the probabilistic approach to power system stability studies.

Case 1. Full loading (100%) of the system with nominal penetration of RESs of 22% and equivalent system inertia of $H = 7.05$ s. The output of some SGs (according to table) is reduced to accommodate RESs. The generators that are chosen to be disconnected (in the NETS and NYPS area only) are assumed to be coal power plants, while the rest of the generators that are neither disconnected nor de-loaded are nuclear and hydro. Gas turbines are de-loaded but not disconnected.

Case 2. Reduced loading (60%) of the system with nominal penetration of RESs of 22% and equivalent system inertia of $H = 7.05$ s. The disconnection of synchronous generation due to RESs is the same as in Case 1. Additional de-loading of all generators is applied equally, to match lower system loading.

Case 3. Reduced loading (40%) of the system with nominal penetration of RESs of 22% and equivalent system inertia of $H = 7.05$ s. The disconnection of synchronous generation due to RESs is the same as in Case 1. Additional de-loading of all generators is applied equally, to match lower system loading.

Case 4. Reduced loading (60%) of the system with nominal penetration of RESs of 37% and equivalent system inertia of $H = 4.23$ s. The rating (and consequently the inertia) of some SGs is reduced to account for lower system loading.

Case 5. Reduced loading (40%) of the system with nominal penetration of RESs of 56% and equivalent system inertia of $H = 2.82$ s. The rating (and consequently the inertia) of some SGs is reduced to account for lower system loading.

Case 6. Reduced loading (60%) of the system with nominal penetration of RESs of 34% and equivalent system inertia of $H = 4.7$ s. Some of the SGs are disconnected as entire power plants.

Case 1 represents the full loading of the system with 22% penetration of renewables, Cases 3 and 5 represent a light loading condition (40%) with a different level of penetration of RESs and different equivalent system inertia, and Cases 2, 4 and 6 represent a medium loading condition with a different level of penetration of RESs and different equivalent system inertia.

The nominal penetration level of RESs is calculated using the following equation:

$$\text{NPL}_a = \frac{\sum_{j=1}^n P_{\text{RES},j}}{\sum_{i=1}^m S_{\text{SG},i} \text{pf}_{\text{SG},i}}, \quad (3.1)$$

where $S_{\text{SG},i}$ is the rated apparent power of individual m SGs in the network, $P_{\text{RES},j}$ is the rated (maximum) power output of n RESs and $\text{pf}_{\text{SG},i}$ is the power factor of SGs. The equivalent system inertia, H_{sys} , of the test system is calculated using (3.2), where, H_i , $S_{\text{SG},i}$ and n represent the inertia constant of each generator, the generator rating and the number of SGs in the system, respectively:

$$H_{\text{sys}} = \frac{\sum_{i=1}^n S_{\text{SG},i} H_i}{\sum_{i=1}^n S_{\text{SG},i}}. \quad (3.2)$$

The daily loading [42] and PV output curves (accounting for different irradiation levels during 24 h) [12] are initially used to sample the hour of the day randomly following a uniform distribution in order to determine the per unit (pu) values for all the loads and all PV units in the network according to the respective curves. For every hour within the day, the corresponding uncertainties are also modelled using a normal distribution for the system load [43] and a beta

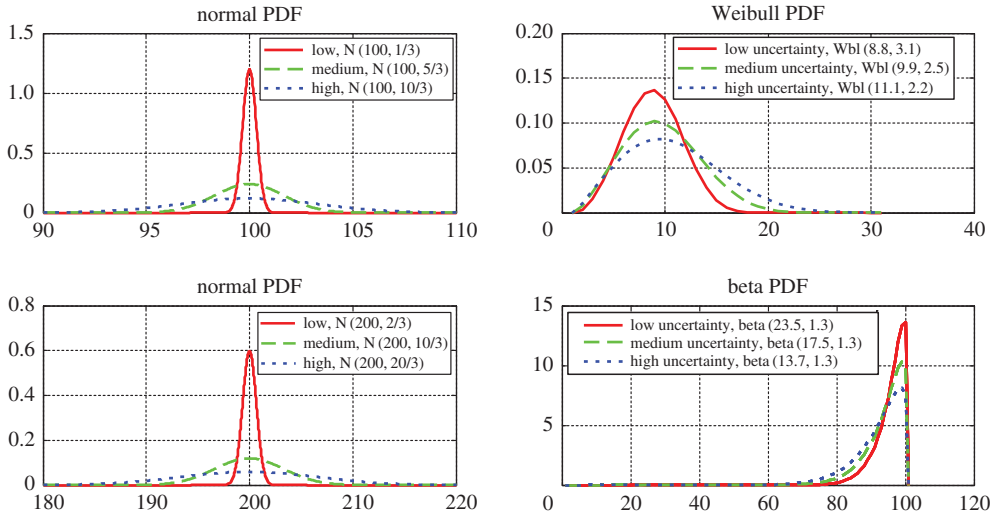


Figure 2. Probability density functions of normal, Weibull and beta distributions considering different levels of uncertainties to represent power generation, wind speed, load and PV output.

distribution for the PV generation [30]. Therefore, an extra uncertainty scaling factor for loads and PVs is introduced, which is then multiplied with the corresponding value from the daily loading or PV curve, respectively. The normal distribution for the system loading uncertainty has a mean value of 1 pu and a standard deviation of 3.33%, and the beta distribution a and b parameters are 13.7 and 1.3, respectively [44]. For wind generation, the mean value of the wind speed within one day is considered constant [45] and the uncertainty of the wind speed is modelled using a Weibull distribution [33]. After considering the wind speed uncertainty, the power curve of a typical wind generator is used [46] to derive the power output. The Weibull distribution parameters used are $\varphi = 11.1$ and $k = 2.2$ [33]. To establish a viable operating point of the system in each case study (starting with predefined initial loading of SGs as defined above for different case studies) after accounting for the uncertainty in loading and RES generation, the distributions are sampled (1000 times in the examples presented in this paper) separately for each load, wind and solar plant, and an optimal load flow solution (OPF) is then used to determine the corresponding output/despatch of SGs. The cost functions for OPF are taken from [43]. The nominal capacity of each generator is then adjusted by adding 15% spare capacity. If the resulting S_{SG} is larger than the nominal apparent power of the generators, it is set to the nominal value and there is no room for disconnection of SGs. Conventional generation disconnection, considering that all SGs in the network represent equivalent generators, is modelled by reducing the nominal power of the generator, which is equivalent to a reduction in the moment of inertia of the power plant and an increase in the generator reactance.

The PDFs corresponding to different parameters of distributions used, representing different levels of uncertainty (low level of uncertainty with $3\sigma = 1\%$ of μ ; medium level of uncertainty with $3\sigma = 5\%$ of μ ; high level of uncertainty with $3\sigma = 10\%$ of μ), are shown in figure 2.

(i) Small disturbance stability

Figures 3 and 4 show the scatter plots of critical modes and cumulative distribution functions (CDFs), respectively, of damping of critical electromechanical modes (the least damped electromechanical modes in the system obtained following appropriate modal analysis) for all six case studies. By inspecting figure 3, it can be determined which of the cases are likely to have a better damped critical mode, i.e. be more stable (modes located further to the left in the complex plane, e.g. Case 2, 60% loading of the systems), and in which of the cases the influence of system

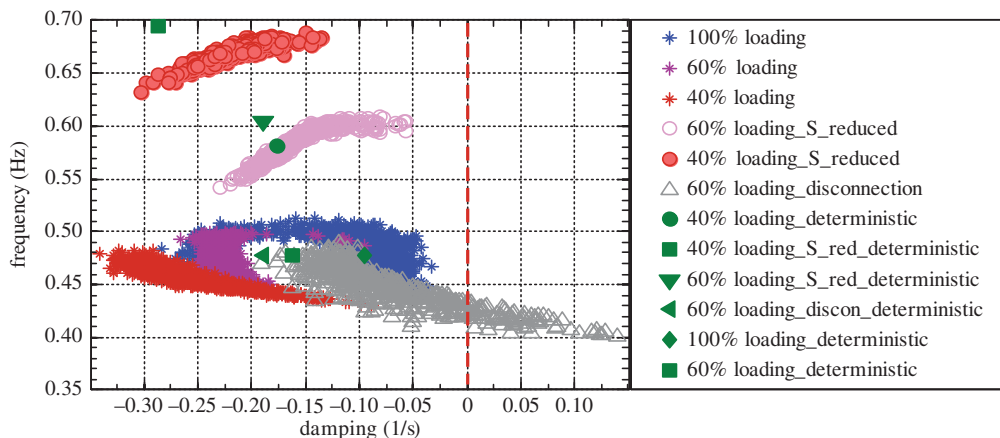


Figure 3. Locations of critical modes in the complex plane due to modelled uncertainties.

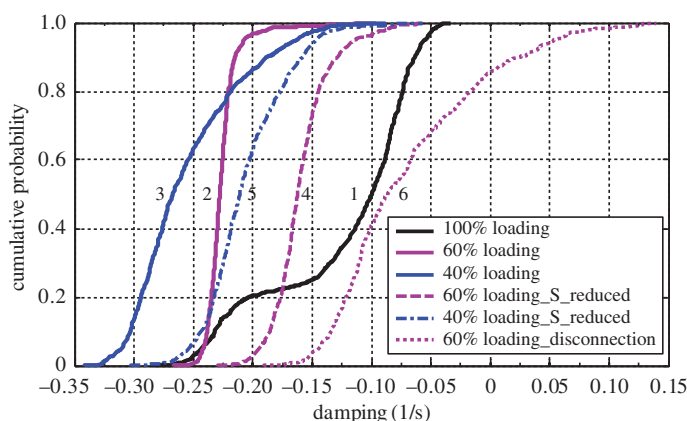


Figure 4. CDF curves of damping of critical modes (case studies 1 to 6, as shown in key and as marked on curves).

uncertainties will have a bigger effect (larger area covered by potential locations of critical modes, i.e. wider spread of eigenvalues, e.g. Case 6, 60% loading with disconnection of some of the SGs) on the critical mode.

Similarly, based on the gradient of the slope of CDFs (figure 4), it can be concluded also that the system uncertainties will have the largest effect in Case 6 and the least in Case 2.

By setting the confidence level of, say, 95%, and inspecting CDFs in figure 4, one can rank the cases in terms of having particular damping of the critical mode, i.e. the rank (from most stable to least stable) of the cases in this example would be 2, 3, 5, 4, 1, 6. In other words, there is 95% probability that the damping of the critical mode in Case 2 will be less than -0.22 , in Case 5 less than -0.15 , in Case 1 less than -0.06 and so on. Note that, due to intersections of CDF curves, a different order would be obtained for a different confidence level, i.e. for a confidence level between 30% and 80% the order would be 3, 2, 5, 4, 1, 6.

From the above illustrative example of case studies performed, it can be seen that the probabilistic small disturbance stability analysis reveals that the damping of the critical mode becomes worse with the increase in system loading and disconnection of SGs, while the frequency of the mode is higher with lower equivalent inertia of the system. The general conclusions from deterministic studies are the same, though deterministic studies cannot identify the range of potential variation in damping and frequency of the critical mode and in some cases may fail to

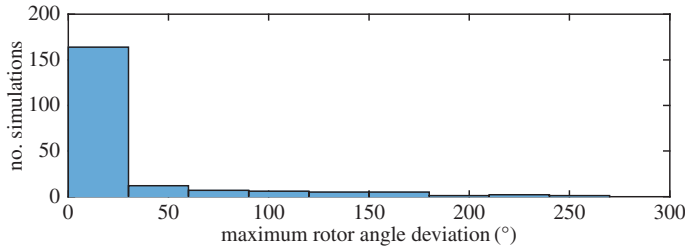


Figure 5. Histogram of maximum rotor angle deviation of G15.

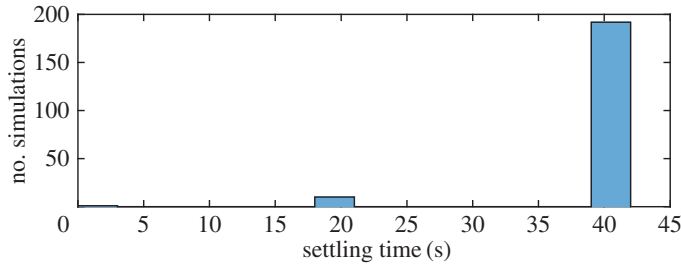


Figure 6. Histogram of settling time of G6.

identify situations when, due to system uncertainties, the system may become unstable to small disturbance (e.g. Case 6).

In general, probabilistic studies can reveal the range of variation of damping and frequency of the critical mode, and consequently the sensitivity of the critical mode to uncertainty in different system parameters, identify the risk of system instability, and provide robust results due to consideration of numerous influential parameters and the range of operating conditions.

(ii) Transient (large) disturbance stability

In studies of transient stability analysis, three phase faults on transmission lines were simulated. The faulted line is selected uniformly among the 66 available lines. The fault position along the length of the line is randomly selected following a uniform distribution. The duration of fault is modelled as a normal distribution as in [47] with a mean value of 13 cycles and a 0.667 cycles standard deviation (2 cycles at 3σ). Although this is a long fault-clearing time for high-voltage systems and only realistic when circuit breakers fail or when delayed tripping is involved [48], it is selected to generate a reasonable mix of stable and unstable conditions in the test system. The faults are assumed to be cleared without tripping any line.

Different indices can be, and have been, used in the past for describing the transient stability status of the system, including maximum generator rotor angle deviation, settling time of rotor oscillations and the transient stability index (TSI). Examples of results of calculation of different stability indices for the critical generator (not necessarily the same generator is identified as critical using a different stability index) obtained in Case 6 are shown in figures 5–8.

The TSI, as one of the most commonly and frequently used indices to describe power system transient stability [49,50], is used in this study for transient stability assessment. The TSI is calculated by

$$\text{TSI} = 100 \times \frac{360 - \delta_{\max}}{360 + \delta_{\max}}, \quad (3.3)$$

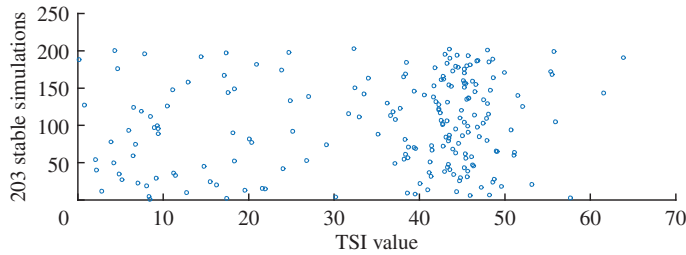


Figure 7. Scatter plot of TSI values of 203 stable simulations (G15). (Online version in colour.)

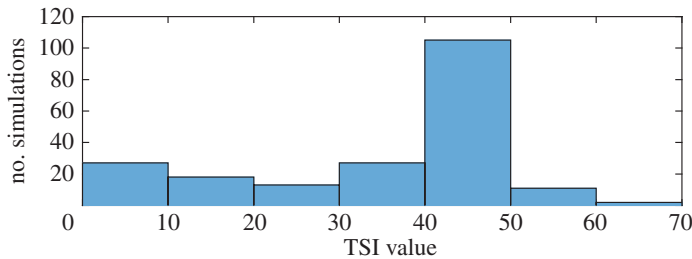


Figure 8. Histogram of TSI values of 203 stable simulations (G15). (Online version in colour.)

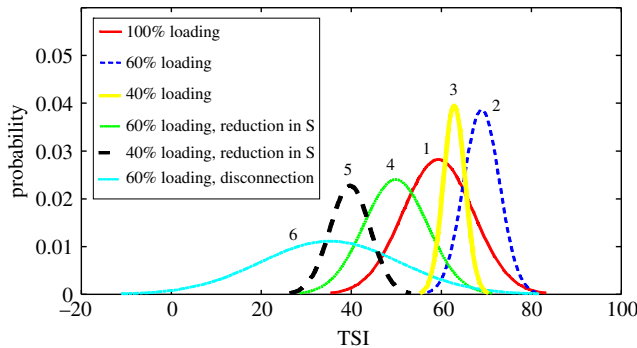


Figure 9. PDFs of TSI for different case studies, as in key and on curves.

where $\delta_{\max}$ is the maximum rotor angle deviation. For a stable system, the TSI is larger than zero, while for an unstable system it becomes negative. The more stable the system is, the larger is the value of TSI, and vice versa.

Figures 9 and 10 show the PDFs and CDFs, respectively, of the TSI values obtained for different case studies. These comparative plots highlight the impact that the loading and disconnection of generators have on the transient stability margin of the power system. Focusing on the CDFs in figure 10, this plot can be used to quickly understand the probability that the TSI will fall below a given level. To facilitate the comparison of the results of different case studies (Cases 1–6), the histograms of TSI obtained in the different case studies (e.g. histogram of figure 8) are represented (fitted) with the most frequently used and best understood normal distribution and such obtained ‘equivalent PDFs’ compared. This is, of course, only a rough approximation, as many of the resulting histograms/distributions cannot be ideally fitted with a normal distribution.

For a more accurate and detailed comparison of results of probabilistic studies, different distributions, appropriate for the observed variation in the chosen index, should be fitted and

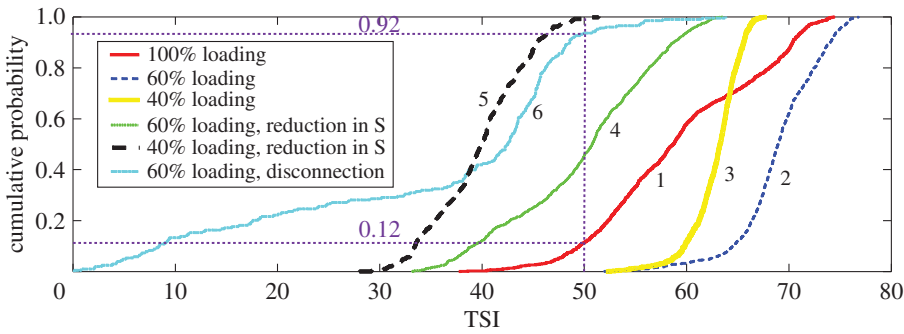


Figure 10. CDFs of TSI for different case studies, as in key and on curves.

compared (e.g. distributions illustrated in figure 2) using appropriate descriptors, moments, of these distributions (e.g. mean, variance, skewness and kurtosis), which are quantitative measures that describe the shape of the dataset [51]. In this paper, a normal distribution is chosen as the illustration and ‘first approximation’ only to show how the results of probabilistic studies can be represented and compared.

As an illustration, consider the probability that the TSI is lower than 50 (given a fault has occurred and the system has remained stable). These numbers can be easily determined from the CDF plot as shown in figure 10. It can be seen that $\Pr(\text{TSI} < 50) \approx 0.12$ for Case 1, $\Pr(\text{TSI} < 50) = 0$ for Cases 2 and 3 and $\Pr(\text{TSI} < 50) \approx 0.92$ for Case 6.

More importantly, the probabilistic approach produces a wealth of probabilistic and statistical data about critical system outputs that can enable a much more advanced understanding of the behaviour of the system. The number of parameters used in the studies must be carefully selected to avoid unnecessarily high computational burden, or efficient sampling methods should be used. This is particularly true for transient stability analysis of large power systems, as it requires dynamic simulations, which can be very time-demanding.

(iii) Frequency stability

In frequency stability studies, the parameters of interest are the frequency nadir (maximum drop in frequency), the rate of change of frequency (ROCOF) and the time of occurrence of a frequency nadir following the disturbance. In each case study considered, the disturbance leading to frequency deviation was simultaneous disconnection of generators G7 and G10. Following the disconnection of these two generators, the frequency response of all remaining generators and the frequency at three tie lines, 43, 44 and 45, were recorded.

The box plot and PDF of frequency nadirs obtained in different case studies are illustrated in figures 11 and 12, respectively. It can be seen from these figures that higher system loading results in a bigger frequency nadir (higher frequency drop), with other parameters kept the same. However, it should be observed that, due to the reduced system loading, the disturbance was also reduced (loading of disconnected generators was lower at the time of disconnection though the same generators were disconnected). By comparing resulting frequency nadirs for Case 3 (reduced loading to 40% with nominal penetration of RESs) and Case 5 (reduced loading to 40% with increased penetration of RESs to 56%, and consequently reduced system inertia) and for Case 2 (reduced loading to 60% with nominal penetration of RESs) and Case 4 (reduced loading to 60% with increased penetration of RESs to 37%, and consequently reduced system inertia), it can be seen that increased penetration of RESs and consequential reduction in system inertia lead to (i) bigger frequency nadirs (the mean value of the corresponding PDF is lower) and (ii) bigger variation in frequency nadirs (wider PDF). The higher penetration of RESs (and consequently lower system inertia), in general, results in a wider spread of frequency nadirs (compare the width of PDFs for Cases 5, 4 and 6).

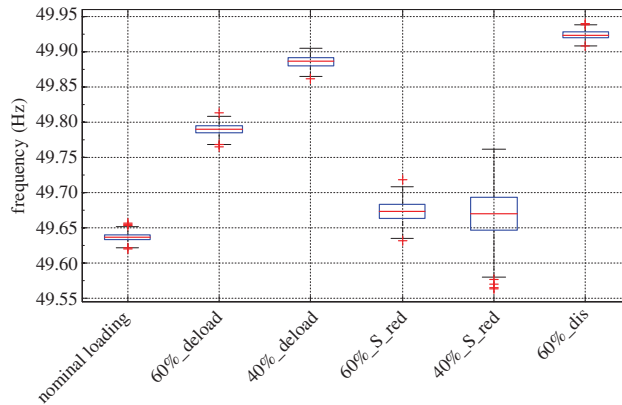


Figure 11. Box plot of a frequency nadir, for Cases 1 to 6 from left. (Online version in colour.)

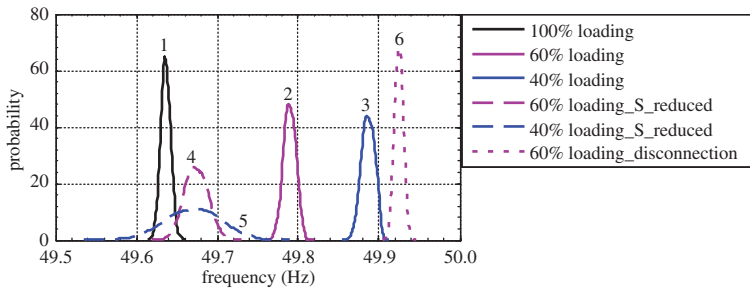


Figure 12. PDF of a frequency nadir, for Cases 1 to 6 as shown. (Online version in colour.)

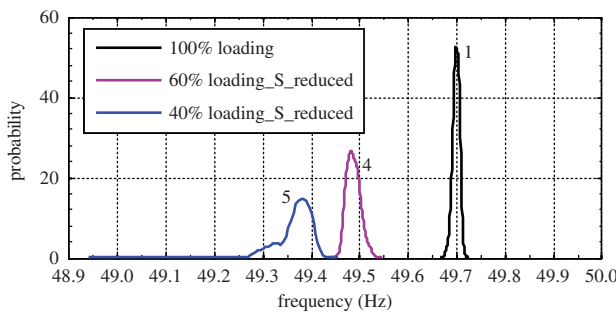


Figure 13. PDF of the frequency nadir for the same size of disturbance, for Cases 1, 4 and 5. (Online version in colour.)

The highest level of penetration of RESs (56%) and the lowest system inertia (2.82 s) in Case 5 lead to the largest variation in frequency nadirs. The results of comparison of frequency nadirs and ROCOF for Cases 1, 4 and 5 (different penetration levels of RESs and different system inertia) are illustrated in figures 13 and 14, respectively. In this case, the size of the disturbance is kept the same in spite of different loading of the system and different penetration levels of RESs. It can be seen from figure 13 that, for the same disturbance, the increase in penetration of RESs and reduction in system inertia lead to higher frequency nadirs and a wider spread of its variation. Similarly, figure 14 shows that the range of variation of ROCOF and its value increase with the level of penetration of RESs (reduction in system inertia).

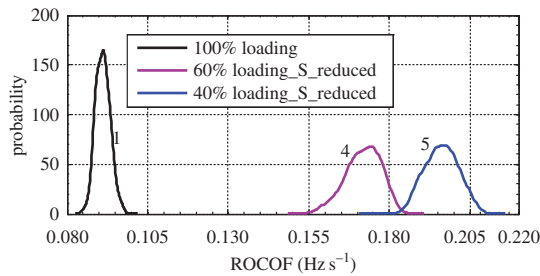


Figure 14. PDF of ROCOF for the same size of disturbance, for Cases 1, 4 and 5. (Online version in colour.)

The results of probabilistic frequency stability analysis demonstrate that the effect of increased system uncertainties becomes amplified in systems with reduced inertia and that the extent of this cannot be judged *a priori* based on the variation of a single system parameter, e.g. penetration of RESs, reduction in system inertia or the system loading level. The deterministic studies, as in the case of other types of stability studies discussed in previous sections, cannot identify the range of variation of different indices depending on system uncertainties nor sensitivity of indices to variation in system uncertainties. Furthermore, the deterministic studies consider only one operating point and a potentially unfavourable combination of system parameters, e.g. the worst-case scenario, the occurrence of which may be extremely low, that could lead to frequency instability. By contrast, through probabilistic studies, the risk of frequency instability and the major parameters contributing to its increase can be calculated, and depending on the value of the risk of instability and possibility to limit the variation of critical parameters, the appropriate decision can be made.

4. Conclusion

The paper discussed the transformative changes in power systems requiring different approaches to power system modelling control and analysis and in particular to power system stability studies. It emphasized the increasing levels of uncertainty in system operation and the sources of these uncertainties as well as the challenges that the operators of these new systems will be facing while ensuring system stability and security of supply. It also presented a brief overview of the main probabilistic approaches that have been used in the past in power system stability studies and demonstrated how these approaches can be used for more informed decision-making regarding system operation.

Finally, the paper showed, using selected illustrative examples and a large power system model with representative uncertainties modelled, how different types of stability studies can be performed and the results of the studies illustrated within the same framework. Conventional deterministic approaches are easy to implement and are less computationally demanding. They, however, cannot identify the worst-case scenario and the likelihood of the system becoming unstable under certain operating conditions. The range of variation of system damping cannot be determined nor can the stochastic nature of operation of RESs be easily incorporated in the analysis. On the other hand, the probabilistic approach requires longer computational times (if efficient sampling techniques and more advanced methods are not used) and more complex mathematical approaches. The complexity of modelling and simulations increases with the use of more sophisticated methods; however, a saving in computational time could be achieved in this way and this renders these approaches particularly suitable for large system studies. The performed analysis highlights the breadth of information that probabilistic studies could offer and that could be used for more informed decisions, making system operation and control better when compared with deterministic analysis. However, one has to be aware of the additional

computational burden that probabilistic analysis may present when applied on large power systems with numerous uncertain parameters.

Different approaches have been proposed in the past to reduce the computational burden of probabilistic studies while maintaining the accuracy and detail of the results, including those mentioned in §2b. The efficient sampling techniques or prior identification of important parameters to be modelled, i.e. ranking of uncertainties in the system based on their influence on results of the performed study, or even parallel computing are therefore mandatory for the probabilistic stability analysis of large power systems. Further information on the computational burden associated with probabilistic studies, the efficient sampling techniques and methods for identification of critical uncertainties to reduce the computational burden of probabilistic studies while preserving the required accuracy can be found in [23,52–55] and associated references.

Data accessibility. Further details on the system, references to data sources used in simulations and appropriate system settings and computational tools can be downloaded from <https://data.mendeley.com/datasets/z8xrbmfg7g/draft?a=213bf123-025e-4154-b0bd-c1ca63685ff2>.

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